

Lab Assignment 2

MTH 308: Numerical Analysis and Scientific Computing–I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: January 23, 2026

This lab will have you implement and understand the effects of cancellation error and truncation error through the numerical examples provided below. Please write a single script file `Lab2LastnameRollNumber.m` and run them all. If you click on the **Publish** button in MATLAB, it will automatically create a folder named `html` containing a file named `Lab2LastnameRollNumber.html` and figures in `.png` format (if any are generated). For example, if my roll number is 260102, my script file should be `Lab2Biswas260102.m`, and the `html` folder should contain the file `Lab1Biswas260102.html` and output figures.

Each solution should:

- Be contained in a separate **code cell** (starting with `%%`) that includes the problem number and a short description.
- Run independently of other cells.
- Be adequately commented to explain your logic.

1. Write a function to solve the quadratic equation $ax^2 + bx + c = 0$ using the standard quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

or the alternative formula

$$x = \frac{2c}{-b \mp \sqrt{b^2 - 4ac}}$$

Your function should accept values for the coefficients a, b , and c as input and produce the two roots of the equation. It should detect when the roots are not real, but need not use complex arithmetic explicitly. It should return a vector $y = [\text{type}, x_1, x_2]$, where type is 0 for real roots and 1 for complex roots. In the case of real roots, x_1 and x_2 should be the two roots, and in the case of complex roots, x_1 and x_2 should be the real and complex parts of the roots respectively. If you get a repeated root, say x , return y such that $x_1 = x_2 = x$.

You should guard against unnecessary overflow, underflow, and cancellation. When should you use each of the two formulas? Try to make your program robust when given unusual input values, such as $a = 0$ or $c = 0$, which otherwise would make one of the formulas fail. Any root that is within the range of the floating-point system should be computed accurately, even if the other is out of range.

Use the template:

```
function y = roots(a,b,c)
- code to calculate y -
end
```

Test your program using the following values for the coefficients:

a	b	c
6	5	-4
6×10^{154}	5×10^{154}	-4×10^{154}
0	1	1
1	-10^5	1
1	-4	3.999999
10^{-155}	-10^{155}	10^{155}

2. (a) Write a program to compute an approximate value for the derivative of a function using the finite-difference formula

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Write your program for the function $\tan(x)$ and use $x = 1$. Use the template:

```
function y = fin_diff_tan(x,h)
```

```
- code to calculate y (the approximate value for the derivative of tan(x)) -  
end
```

Determine the error by comparing your result with the exact derivative, $\sec^2(x)$, using the MATLAB built-in function. Use the template:

```
function e = error_fd_tan(x,h)
```

```
- code to calculate e (error in approximate value of derivative of tan(x)) -  
end
```

Plot the magnitude of the error as a function of h , for $h = 10^{-k}$, $k = 0, \dots, 16$. You should use log scale for h and for the magnitude of the error. Use the template:

```
function [] = plot_error(x,K)
```

```
- code to plot absolute error against h on the log scale, where k = 0:K -  
end
```

Is there a minimum value for the magnitude of the error? How does the corresponding value for h compare with the rule of thumb $h \approx \sqrt{\epsilon_{mach}}$?

- (b) Repeat the exercise using the centered difference approximation

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

Define the functions `cent_diff_tan(x,h)` and `error_cd_tan(x,h)` similarly.

Include this second error curve on the same plot for comparison (add it to the `plot_error` function).

- (c) Observe the linear portions of the two error curves (the parts where the error decreases as h decreases). Estimate the slopes of these linear segments on the log-log plot. What do these different slopes tell you about how the accuracy of each method improves as you reduce the step size h ?